

API Parameters:

1. model:

- **What it does:** Specifies which "brain" you want to use.
- **Example:**
 - `model="gpt-4o-mini"` → Fast, cheap, simpler answers.
 - `model="gpt-4o"` → Slower, expensive, highly complex reasoning.

2. messages (or prompt or content):

- **What it does:** The array containing the conversation history and system rules.

Ex.

```
"messages": [  
  {"role": "system", "content": "You are a helpful assistant."},  
  {"role": "user", "content": "Hi, my name is John."}  
]
```

The AI model replies: "Hello John, how can I help you?"

```
"messages": [{"role": "user", "content": "What is my name?"}]
```

If you do, the **AI will say** "I don't know."

Instead, you must send the **entire history** in the messages parameter:

```
"messages": [  
  {"role": "system", "content": "You are a helpful assistant."},  
  {"role": "user", "content": "Hi, my name is John."},  
  {"role": "assistant", "content": "Hello John, how can I help you?"},  
  {"role": "user", "content": "What is my name?"}  
]
```

*The AI model looks at the whole list, sees the past context, and **replies**: "Your name is John."*

for roles refer: - Roles sheet*

3. max_tokens (or max_output_tokens)

- **What it does:** The absolute maximum number of words/tokens the AI model is allowed to generate.

- **Example Prompt:** "Explain quantum physics."
 - **max_tokens=5** → Output: "Quantum physics is the study" (Cuts off abruptly)
 - **max_tokens=500** → Output: (Full 3 paragraph explanation)

4. stop (or stop_sequences)

- **What it does:** A list of words or characters. If the AI model generates them, it stops immediately.
- **Example Prompt:** "Count to 10."
 - **No stop sequence** → Output: "1, 2, 3, 4, 5, 6, 7, 8, 9, 10."
 - **stop=["5"]** → Output: "1, 2, 3, 4, " (Stops before printing 5)

5. response_format

- **What it does:** Forces the AI to output in a specific structure (usually JSON).
- **Example Prompt:** "Name two planets."
 - **Default (text)** → Output: "Two planets are Mars and Venus."
 - **{"type": "json_object"}** → Output: {"planets": ["Mars", "Venus"]}

6. temperature

- **What it does:** Controls randomness/creativity (usually 0.0 to 2.0).
- **Example Prompt:** "Write a sentence about a cat."
 - **temperature=0.1 (Low)** → Output: "The cat sat on the mat." (Safe, predictable)
 - **temperature=1.2 (High)** → Output: "The ginger ninja pounced on the neon yarn!" (Creative, varied)

for temperature refer: - temperature sheet*

7. reasoning_effort / thinking_budget / thinking effort:

- **What it does:** Controls how much hidden "thinking" the model does before answering.
- **Example Prompt:** "What is 17×24 ?"
 - **low effort** → Thinks for 1 sec → Answers "408" (Quick math)
 - **high effort** → Thinks for 10 secs → Answers "408" (Calculates it 3 different ways internally to verify)

for thinking effort refer: - thinking effort sheet*

8. Top-K (The "Count" Filter):

What it does: It tells the AI model to only look at the **top K number of words on the list**, regardless of their probability.

- **prompt:** "Only pick from the **Top 3** most popular items on the menu."

Example (If K = 3): The AI model looks at the list and keeps exactly 3 words.

1. Apple (60%) - Kept
2. Banana (30%) - Kept
3. Cherry (8%) - Kept
4. Car (1%) - *Discarded*
5. Zebra (0.001%) - *Discarded*

9. Top-P (The "Percentage" Filter):

What it does: Also known as *Nucleus Sampling*. It tells the AI model to keep adding words from the top of the list *until* their **combined probabilities hit a certain percentage (P)**.

- **prompt:** "Give me the most popular menu items that make up 90% of all customer orders."

Example (If P = 0.90 / 90%): The AI model starts at the top and adds up the percentages until it hits 90%.

1. Apple (60%) - Kept (*Total: 60%*)
2. Banana (30%) - Kept (*Total: 90% STOP! We hit 90%. Everything else is thrown away.*)
3. Cherry (8%) - *Discarded*
4. Car (1%) - *Discarded*

Note:

- Top-K limits by a hard number of choices.
- Top-P limits by a target percentage of probability.

10. Seed

- **What it does:** By default, if you ask an AI model the same question twice, you will get two slightly different answers because of the randomness (Temperature/Top-P). The seed parameter lets you lock in that randomness by providing a specific integer number.

- If you use the exact same prompt, same temperature, and the same seed number, the LLM will generate the exact same answer every time.

Example:

Attempt 1:

Prompt: "Write a haiku about a cat."

Seed: 42

Output: "Furry little paws / Sleeping in the sun all day / Purring softly now"

Attempt 2:

Prompt: "Write a haiku about a cat."

Seed: 999 (Different seed = different random path)

Output: "Chasing laser lights / Jumping high upon the bed / Master of the house"

Attempt 3:

Prompt: "Write a haiku about a cat."

Seed: 42 (Same seed as Attempt 1 = identical output)

Output: "Furry little paws / Sleeping in the sun all day / Purring softly now"

11. frequency_penalty:

- **What it does:** Penalizes the AI model for repeating the exact same words. Forces it to use synonyms.
- **Example Prompt:** "Describe a happy dog."
 - **o.o (None)** → Output: "The happy dog was very happy to see his owner. The happy dog wagged his tail."

- **1.5 (High)** → Output: "The joyful canine was thrilled to greet his human. The ecstatic pup wagged his tail."

12. presence_penalty

- **What it does:** Penalizes the AI for sticking to the same topic. Forces it to bring up new ideas.
- **Example Prompt:** "Give me advice for staying healthy."
 - **0.0 (None)** → Output: "Eat vegetables. Vegetables have vitamins. Carrots are good vegetables." (Stuck on vegetables)
 - **1.5 (High)** → Output: "Eat vegetables. Also, ensure you sleep 8 hours. Drink plenty of water and exercise." (Moves to new topics)

13. logit_bias

- **What it does:** Mathematically bans or forces specific words by targeting their internal Token ID.
- **Example Prompt:** "Say hello."
 - **No bias(preference)** → Output: "Hello! How can I help you?"
 - **{"hello_token_id": -100} (Banned)** → Output: "Greetings! How can I assist?" (It is physically blocked from using the word "Hello").

14. Logprobs (log probabilities)

- **What it does:** Returns the mathematical probability score for every word it typed.

Example:

User prompt: which is capital of karnataka

output:

WITHOUT logprobs:

```
{
  "content": "Bengaluru"
}
```

WITH logprobs:

```
{
  "content": "Bengaluru",
  "logprobs": [
    {
      "word": "Bengaluru",
      "confidence": 99.8%,
      "other_options": [
        {"word": "Bangalore", "confidence": 0.15%},
        {"word": "Blore", "confidence": 0.04%}
      ]
    }
  ]
}
```

15. tools (or functions)

- **What it does:** You give the AI a list of tools it can use (like a calculator or web search).
AI models use tool/s to communicate with the outside world like other systems/services, calculators, ..

- **Example:**

User prompt::

```
{ "content": "What's the weather in Tokyo?", "tool": "get_weather_api" }
```

AI model call to weather api → { "call_tool": "get_weather", "location": "Tokyo" }

Tool response (from API) to the AI model → { "tool_output": { "condition": "sunny",
"temperature_c": 32 } }

AI model response to user after fetching data from api

```
{ "content": "sunny" }
```